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LAB 0: Remote Server & GPU Setup

COE 201 - AI Large Models, Spring 2026

2026-04-20

College of Engineering, SUSTech

Office Hour: Thursday, 14:30-15:30 @ 3rd Research Building, Room 610

Outline

SSH Connection

Basics of SSH

SSH (Secure Shell) is the standard for connecting to remote Linux machines.

```shell
ssh <username>@<server-address>
# Example: ssh student@10.20.10.1
```

Key Concepts:

- **Host**: The IP address or domain name.
- **User**: Your account on the remote machine.
- **Port**: Usually 22 (default).
- **Authentication**: Password (default) or SSH Key (recommended).

SSH Key Authentication (Passwordless Login)

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Stop typing passwords every time!

1. **Generate Keys (on your local machine):**

```shell
ssh-keygen -t ed25519
```

2. **Copy to Server:**

```shell
ssh-copy-id <username>@<server-address>
```

Now you can log in without a password securely.

On Windows, check this out: https://www.purdue.edu/science/scienceit/ssh-keys-windows.html (The OpenSSH client is installed by default on Windows 10/11, but you may still need to enable it manually following the instruction.)

SSH Config for Productivity

Manage multiple servers easily by editing `~/.ssh/config` on your **local machine**:

```
Host coe201-gpu-lab
    HostName 10.20.10.1
    User student
    Port 22
    IdentityFile ~/.ssh/id_ed25519
```

Usage: Just type `ssh coe201-gpu-lab`!

Pro-tip: VS Code's **Remote-SSH** extension uses this config to give you a full IDE experience on the server.

Copying Files: scp & rsync

How to move your local code or data to the remote server.

Using `scp` (Simple Copy):

```shell
# Copy local file to server
scp ./dataset.zip coe201-gpu-lab:~/workspace/
```

Using `rsync` (Recommended): Only syncs changed files.

```shell
# Sync local folder to remote (note the trailing slashes)
rsync -avzP ./my-project/ coe201-gpu-lab:~/workspace/my-project/
```

- `-a`: Archive mode.
- `-v`: Verbose (shows details).
- `-z`: Compress data during transfer.
- `-P`: Show progress and allow resuming.

Pro-tip: VS Code's file explorer makes this drag-and-drop.

Linux Workspace

Essential Commands

You are in a shell (likely `bash` or `zsh`).

Category	Commands
Navigation	`pwd`, `ls`, `cd <dir>`
Files	`mkdir`, `touch`, `cp`, `mv`, `rm`
Content	`cat`, `head`, `tail`, `grep`
Permissions	`chmod`, `chown`
Editors	`vim`, `nano`, `code` (VS Code Remote)

Tips: Install `tldr` from https://github.com/tldr-pages/tldr and use like this:

- `tldr ls`, `tldr vim`, `tldr rsync`, etc.
- Give you a short and very useful description of commands.

Persistence with tmux

If you close your laptop or the Wi-Fi drops, your SSH connection dies and so does your script.

Solution: Terminal Multiplexers (`tmux`)

- Keeps sessions alive even if you disconnect.
- Allows split panes (left for code, right for monitoring).

Basic Workflow:

1. `tmux new -s training` - Create a session.
2. `Ctrl + b`, then `d` - Detach (the session keeps running).
3. `tmux attach -t training` - Re-attach anytime.

Dev Environment

uv on Linux

Modern Python management works great on servers.

Installation:

```shell
curl -LsSf https://astral.sh/uv/install.sh | sh
```

Project Setup:

1. Clone your repo, with `pyproject.toml`.
2. `uv sync` - Installs everything in a local `.venv`.
3. `uv run python script.py` - Runs your code in the isolated environment.

Alternative: Conda Environment

Conda is a powerful tool for managing both Python environments and non-Python dependencies (like CUDA/CUDNN).

Basic Commands:

- **Create:** `conda create -n coe201 python=3.10`
- **Activate:** `conda activate coe201`
- **List:** `conda env list`
- **Deactivate:** `conda deactivate`
- **Add packages:** `conda install numpy pytorch torchvision`
- **Remove:** `conda env remove -n coe201`

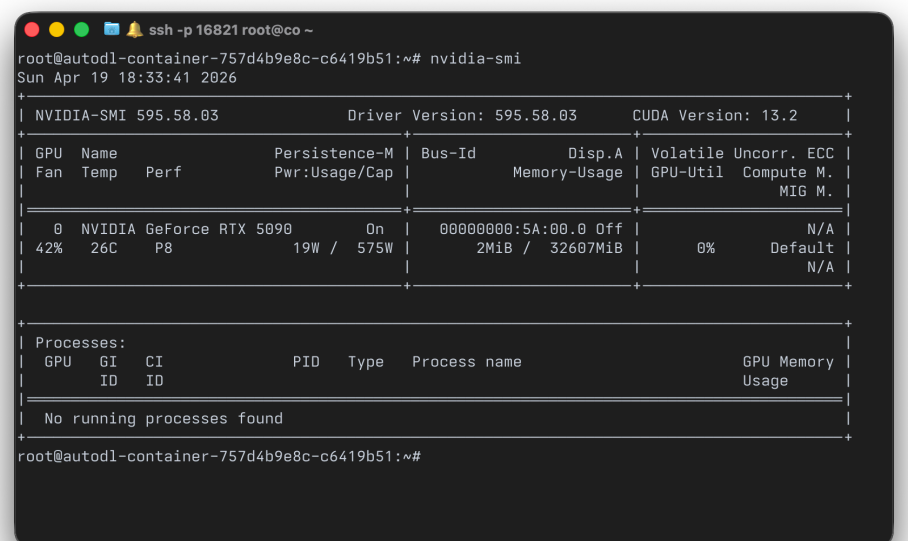
Note: If `conda` is not found, you may need to install **Miniconda** or run `conda init`.

GPU Monitoring

nvidia-smi

The standard tool for NVIDIA GPUs.

```shell
nvidia-smi
```



What to watch:

- **GPU-Util**: How busy the compute cores are (0% to 100%).
- **Memory-Usage**: How much VRAM is used (Critical for LLMs!).

Useful when you share a GPU server with others. Check before running your code.

Live Monitoring

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Auto-refresh nvidia-smi:

```shell
watch -n 1 nvidia-smi
```

nvtop (for a better UI, if installed):

```shell
nvtop
```

Memory Leak Warning: If your script crashes with "Out of Memory" (OOM), check if old processes are still hanging. Use `kill -9 <PID>` to clear them if necessary.

Lab Tasks

Lab 0 Exercises (Counted into Lab 8 Score)

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Task 1: Server Connectivity [10 points]

- Log in to your assigned GPU server using SSH.
- Configure SSH config (`~/.ssh/config`) for a simple login alias.

Task 2: Environment Checkup (`gpu_check.py`) [10 points]

- Navigate to `labs/lab0`.
- Install dependencies and run `gpu_check.py`.

Task 3: tmux Persistence [10 points]

- Run the check script inside a `tmux` session.
- Practice detaching and re-attaching to maintain the process.

Happy Coding on GPUs!